RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Department of Mechanical Engineering

Date: 3rd December 2024	**CIE – II**	**Max. Marks: 50 + 10**
Semester: III	**UG**	**Duration: 120 Min**
Course Title: Material Science for Engineers		**Course Code: ME232TB**

S No	Questions	M	BT	CO
	Part A (Quiz)			
1.	Semiconductor device with a p-n junction which produces current by absorption of light in the depletion region is termed as _____.	1	1	2
2.	_____ is the metal joining process in which parent metal pieces are heated either to plastic or molten state.	1	1	2
3.	If the emission of radiation occurs longer than 10^{-8} seconds is known as _____.	1	1	2
4.	Alloying elements of HSS are _____.	1	1	2
5.	Lightest of commonly used metals is _____.	1	1	2
6.	Define young's modulus of a material with equation.	2	2	2
7.	_____ is the type of glass that cannot allow heat through it because of air between the layers and acts as good insulators.	1	1	2
8.	Concrete is a composite material made up of _____ and _____.	2	1	2
	Part B (Test)			
1a	What is piezoelectric effect? Explain the applications of piezoelectric materials.	05	2	2
1b	Distinguish between ductile and brittle fracture.	05	2	2
2	With the stress strain diagram explain the mechanical properties of a ductile metal.	10	2	2
3a	Discuss the applications of semiconductors.	05	3	2
3b	Explain four different types of cast iron based on microstructure, properties and applications.	05	2	2
4a	With a neat sketch describe the steps to be followed in metal casting process.	05	2	2
4b	Compare thermoplastics with thermosetting plastics.	05	3	2
5a	Justify the need for composite materials. Classify composite materials.	05	3	2
5b	What is a biomaterial? List the desirable properties of biomaterials.	05	2	2

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Department of Mechanical Engineering

Date: 20th February 2024	**CIE – II**	**Max. Marks: 50**
Semester: 3	**UG**	**Duration: 90 Minutes**
Course Title: Materials Science for Engineers		**Course Code: ME232AT**

S No	Questions	M	BT	CO
1a.	With the help of a stress-strain diagram explain toughness and ductility of a material.	04	3	2
1b.	Using stress-strain diagram distinguish ductile and brittle fracture.	06	2	2
2.	Explain the following properties: thermal expansion coefficient, thermocouple, luminescence, optical fibres.	10	3	2
3.	Describe the following: ferroelectricity, piezoelectricity, superconductor.	10	4	2
4.	Explain the properties and applications of following materials with examples. Ceramics and Fiber reinforced composites.	10	3	2
5.	Explain the properties and applications of following materials: stainless steel, cast iron and titanium alloys.	10	3	2

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Department of Civil Engineering

Date: 3/12/2024	**CIE – II (Quiz + Test)**	**Max. Marks: 60**
Semester: III	**UG**	**Duration: 90 Min + 20 Min**
Course Title: Environment and Sustainability (Basket Course)		**Course Code: CV232AT**

S No	Questions	M	BT	CO
	Part A			
1.	Describe the term of "environmental audit"?	1	1	2
2.	Explain the need for new energy sources.	1	2	3
3.	Classify different types of energy sources.	1	2	3
4.	Differentiate between linear and cyclical resource management systems.	1	2	3
5.	Describe the working principle of tidal energy converter in renewable energy generation.	2	1	2
6.	State the significance of GDP in environmental economics.	2	1	2
7.	Enumerate the importance of environmental economics.	2	2	3
	Part B			
1a.	Illustrate carbon cycle with sketch.	05	2	2
1b.	What are the advantages of hydrogen energy?	05	2	2
2.	Describe ocean thermal energy conversion with its advantages and limitations.	10	2	2
3.	Enumerate the concept of sustainable urbanization and the need for the same.	10	2	3
4a.	Discuss the steps in environmental audit.	05	2	3
4b.	Enumerate the economics and social aspects of sustainability.	05	2	3
5.	Illustrate the concept of economics and its integration to social aspects for sustainability.	10	2	3

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Department of Civil Engineering

Date: 20-02-2024	**CIE – II (Test)**	**Max. Marks: 50**
Semester: III (Basket Course)	**UG**	**Duration: 90 Min**
Course Title: Environment and Sustainability		**Course Code: CV232AT**

S No	Questions	M	BT	CO
1.	Enumerate the process of energy management and energy conservation with its objectives.	10	2	3
2.	Explain any 5 types of alternate energy sources with their advantages and disadvantages.	10	3	1
3.	Enumerate the objectives of sustainable urban projects.	10	2	3
4.	Highlight five important principles of sustainable development.	10	3	3
5a.	Briefly explain any 5 international protocols and agreements that address specific environmental issues.	05	2	2
5b.	Describe any 5 disadvantages of linear resource management systems.	05	3	2

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Department of Biotechnology

Date: December 3, 2024	**Closed Book Offline Quiz and Test – 2**	**Max. Marks: 10 + 50**
Semester: III	**UG**	**Duration: 120 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT232AT**

S No	Questions	M	BT	CO
	Part A			
1	A passenger in flight found a bolt and nut in the burger. Identify the type of the hazard.	1	2	3
2	Expand the term GMP.	1	2	4
3	Name any one food certification followed in the world.	1	2	3
4	Expand the term GHP.	1	2	4
5	A person found a frog leg in the food. Identify the type of the hazard.	1	2	4
6	_____ types of licences issued by FSSAI.	1	2	3
7	Name two diseases caused due to food contamination.	1	2	4
8	Milk is heated to _____ °C temperature during pasteurization.	1	1	3
9	What is 2-hour 4-hour rule for food storage.	2	2	3
	Part B (Test)			
1	Define food contamination. Mention and explain with examples the types of Hazard which contaminate food.	10	2	3
2	Elaborate on the basic rules of hygiene, sanitation and safety in food processing.	10	3	3
3	Explain in detail how to eliminate the different type of hazards which contaminate food.	10	3	3
4a	Mention the principles of HACCP.	5	3	4
4b	Decode the FSSAI license number 20719078324011 (Lic. No. as printed on the FSSAI logo).	5	2	3
5a	Explain in detail the instruments used in food testing laboratory.	5	3	4
5b	Illustrate the various analytical methods used in food testing process.	5	3	4

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Department of Biotechnology

Date: February 20, 2024	**Closed Book Offline Test – 2**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 min**
Course Title: Bio Safety Standards and Ethics (Basket course)		**Course Code: BT232AT**

S No	Questions (Test)	M	BT	CO
1.	Explain briefly the importance of The Cartagena Protocol on Biosafety.	10	2	3
2.	Mention the physical, chemical, and biological hazardous material which contaminate the food.	10	2	3
3.	List out various methods to eliminate the chemical, Biological and physical hazardous materials from food.	10	3	3
4.	Expand the terms FSSAI & HACCP. Elaborate on the seven principles of HACCP.	10	3	4
5a.	Illustrate on various Techniques that take place in a food testing laboratory.	5	4	4
5b.	Mention and briefly explain the five methods used for food analysis applicable to samples.	5	4	4